

HAT-P-18b

R-2420

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-18_200708 · created 2026-08-21T23:37:53+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459038.82571 +/- 0.00013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.107 +/- 0.042
Transit depth (Rp/Rs)^2	1.14 +/- 0.9 [%]
Orbital Inclination (inc)	88.77 +/- 1.73
Airmass coefficient 1 (a1)	1.283 +/- 0.027
Airmass coefficient 2 (a2)	-0.174 +/- 0.01
Scatter in the residuals of the lightcurve fit is	2.06 %
Best Comparison Star	None
Optimal Aperture	4.03
Optimal Annulus	12.54
Transit Duration (day)	0.097 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.107 ± 0.042 vs 0.1356 (deviation 0.57σ)
Duration fit vs published	0.097 d vs 0.1129 d (deviation 0.45σ)
Mid-time O–C	-0.3 min
Depth SNR	2.1
Residual scatter	2.06 %

Light curve

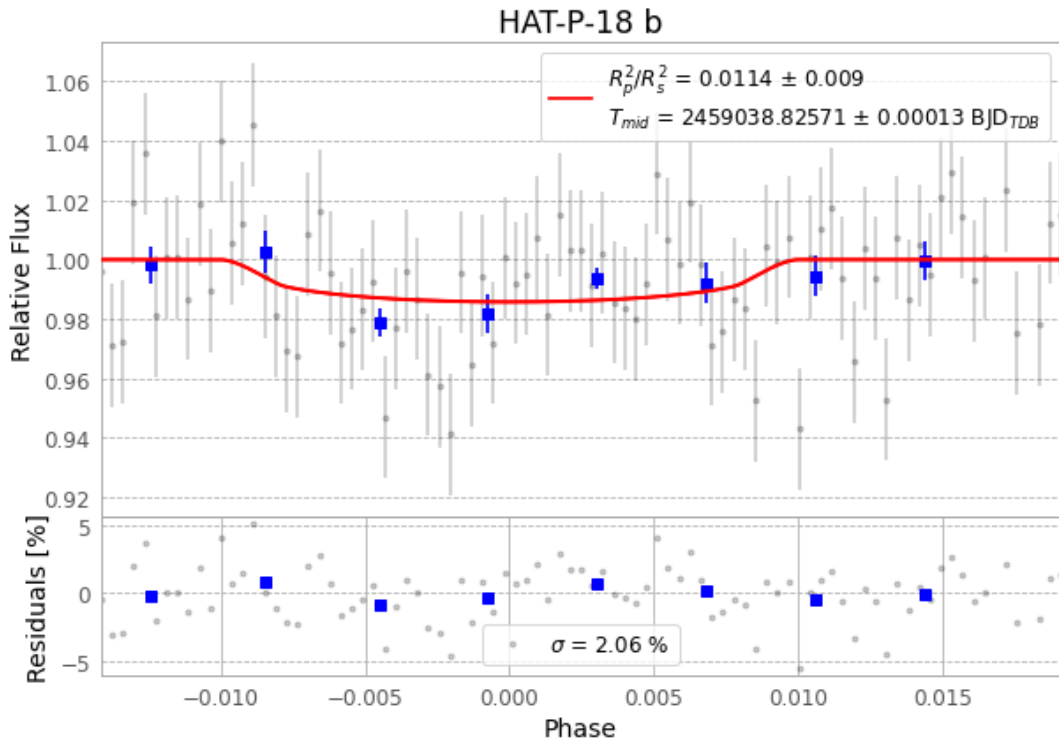


Figure 1 — FinalLightCurve_HAT-P-18 b_2020-07-07.png (SHA-256 41e6dc0ed5313de4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [217, 224], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 309c811e92290d4d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

89 FITS frames (58 MB), HATP-18200708055112.FITS → HATP-18200708101512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-18-200708.log (7bff2e08d0c0...), FinalLightCurve_HAT-P-18 b_2020-07-07.png (41e6dc0ed531...), FinalLightCurve_HAT-P-18 b_2020-07-07.csv (fa438a7ddcec...)

Compute resources

Wall time 118.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 23:37Z.